Introduction to Machine Learning

1. What is Machine Learning?

Machine Learning is a branch of artificial intelligence in which computers learn patterns from data and use those patterns to make predictions or decisions. The major types of machine learning are supervised learning, unsupervised learning, and reinforcement learning.

2. Supervised Learning

In supervised learning, a model learns from labelled training data. Each training example contains input features and a target output. Common supervised learning algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, and K-nearest neighbours.

3. K-Nearest Neighbours (KNN)

KNN is a simple supervised learning algorithm used for classification and regression. For a new observation, KNN calculates its distance from training observations, selects the K closest observations, and uses their values to make a prediction. For classification, the most common class among the neighbours is usually selected.

4. Choosing the Value of K

The value of K strongly affects KNN performance. A very small K, such as K=1, can make the model sensitive to noise and may lead to overfitting. A very large K can make the model too smooth and may cause underfitting. A common approach is to test several K values using validation data or cross-validation and select the value that gives the best validation performance. The selected K should balance bias and variance.

5. Distance Measures

KNN commonly uses Euclidean distance. For two points x and y, Euclidean distance is calculated as the square root of the sum of squared differences between corresponding features. Other distance measures can also be used depending on the data.

6. Feature Scaling

Feature scaling is important for KNN because distance calculations can be dominated by features with larger numerical ranges. Standardization and min-max normalization are common preprocessing techniques.

7. Model Evaluation

Classification models can be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. Regression models can be evaluated using metrics such as mean absolute error, mean squared error, root mean squared error, and R-squared.

8. Summary

Machine learning enables systems to learn from data. KNN is an intuitive example of supervised learning. For KNN, choosing an appropriate K is important: small K values can overfit, while large K values can underfit. Cross-validation is a reliable way to compare candidate K values and select an appropriate value.